College Algebra Test 3 Review

1) Solve the system of equations.

$x + y = 10$

$x - y = 6$

2) Solve the system of equations.

$x + 2y = 12$

$x - 2y = 16$

3) Solve the system of equations.

$4x + y = 5$

$4x + y = 7$

4) Solve the system of equations.

$2x + y = 3$

$6x + 3y = 7$

5) Solve the system of equations.

$x + y = 3$

$3x + 3y = 9$

6) Solve the system of equations.

$-2x + 3y + 4z = 10$

$x + 2y + 3z = 30$

$5x + 4y + z = 13$

7) Solve the system of equations.

$$x + z = 12$$
$$-y + z = 30$$
$$12x + 13y + z = 8$$

8) Solve the system of equations.

$$y + 2z = 8$$
$$-3y + 3z = 10$$
$$2x + 4y + 5z = 9$$

9) Multiplication of Matrices

$$\text{Multiply} \begin{bmatrix} 1 & 5 & 4 \\ 7 & 8 & 11 \\ 2 & 9 & 15 \end{bmatrix} \cdot \begin{bmatrix} -2 & 15 & 6 \\ -4 & 2 & 1 \\ -3 & 7 & 5 \end{bmatrix}$$

10) Multiplication of Matrices

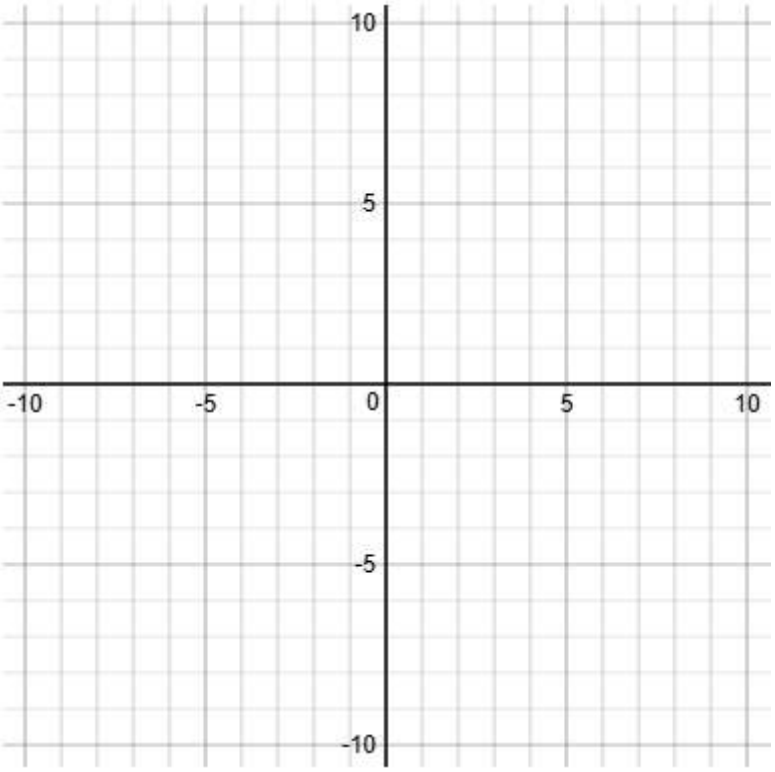
$$\text{Multiply} \begin{bmatrix} -2 & 1 & -3 \\ 1 & 8 & 1 \\ 6 & 11 & 5 \end{bmatrix} \cdot \begin{bmatrix} 2 & 5 & 2 \\ -3 & 2 & 1 \\ 6 & 9 & 8 \end{bmatrix}$$

11) Multiply by a constant: $\frac{1}{2} \cdot \begin{bmatrix} -2 & 15 & 6 \\ -4 & 2 & 1 \\ -3 & 7 & 5 \end{bmatrix}$

12) Multiply by a constant: $4 \cdot \begin{bmatrix} 1 & 5 & 4 \\ 7 & 8 & 11 \\ 2 & 9 & 15 \end{bmatrix}$

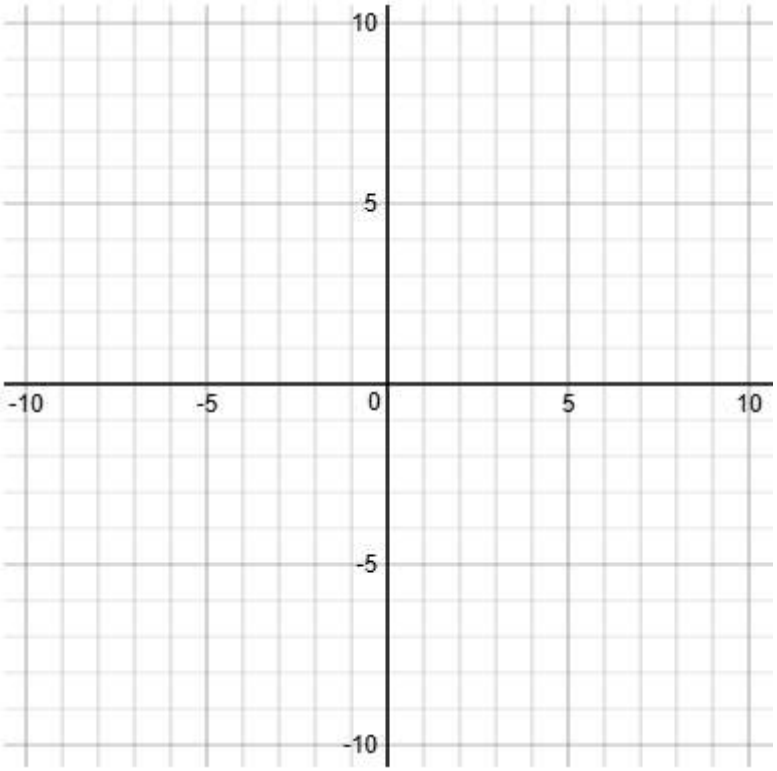
13) Graph the solution set of the linear inequality in two variables.

$y \geq 3x - 1$



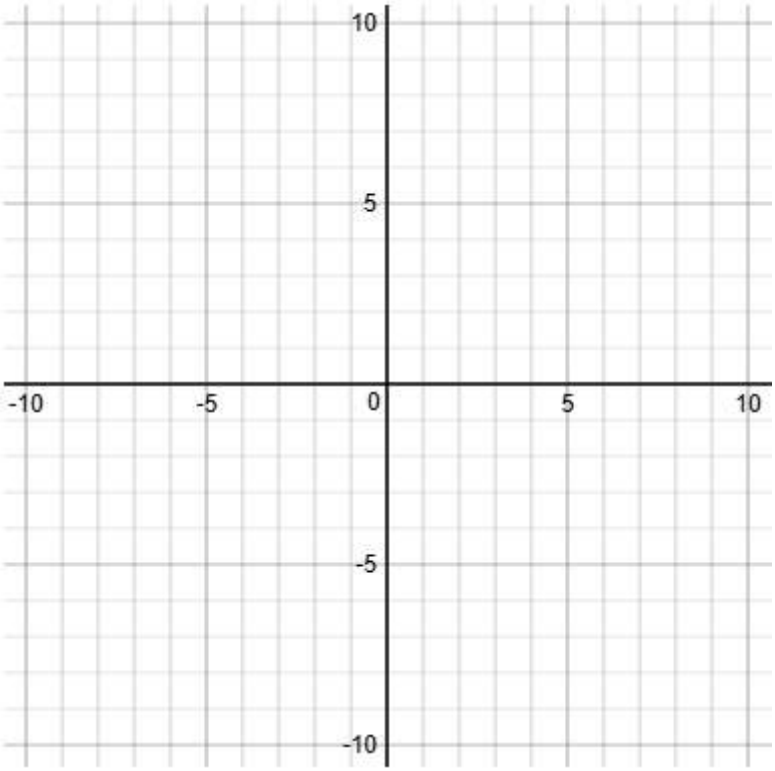
14) Graph the solution set of the linear inequality in two variables.

$y \leq \frac{2}{3}x + 2$



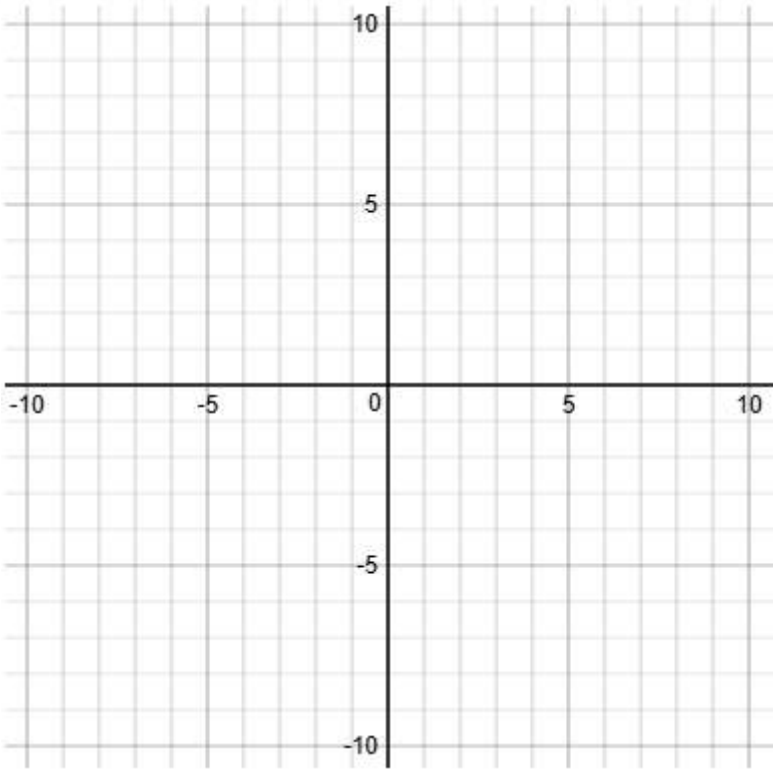
15) Graph the solution set of the linear inequality in two variables.

$3x - 4y \geq -12$



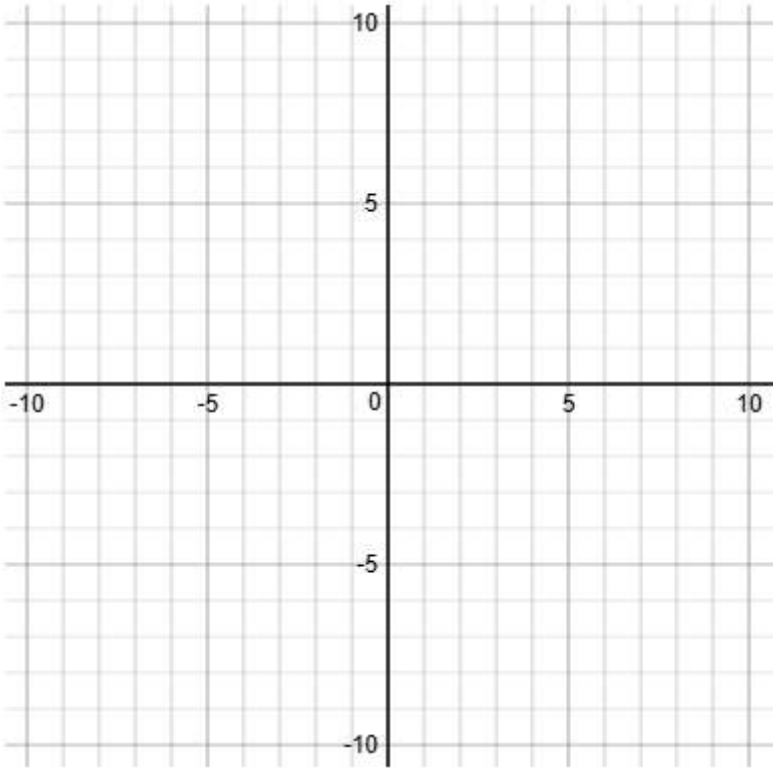
16) Graph the solution set of the linear inequality in two variables.

$2x - 5y < -10$



17) Graph the solution set of each system of inequalities.

$y \leq 2x - 1$ and $y \geq -3x + 2$



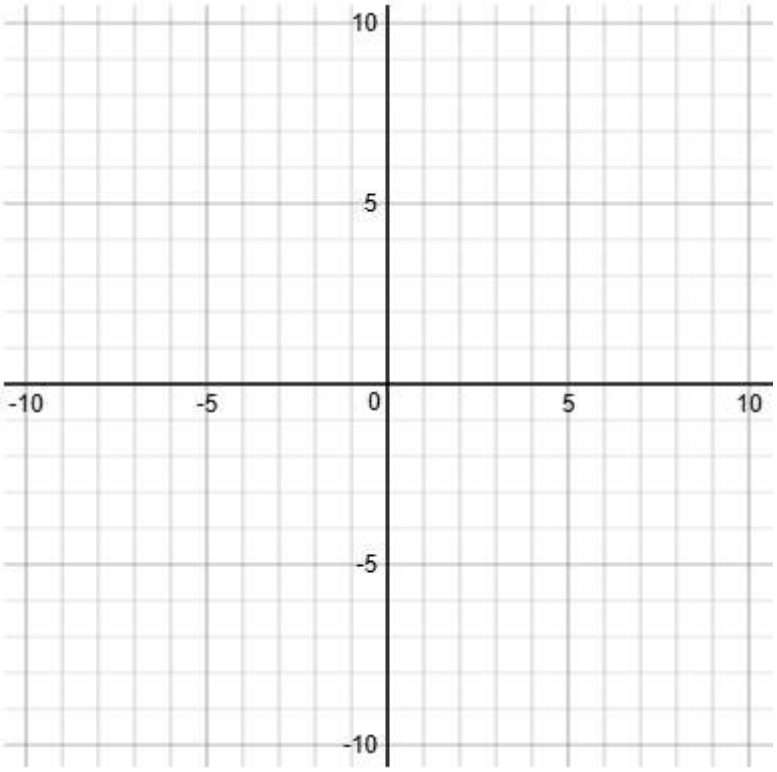
18) Graph the solution set of each system of inequalities.

$2x + 3y \leq 6$ and $5x - 2y \geq 10$

a) Solve the inequality $2x + 3y \leq 6$ for y. $y \leq$ ___?___

b) Solve the inequality $5x - 2y \geq 10$ for y. $y \leq$ ___?___

c) Graph the solution set of the system of inequalities.
(Use the results from part (a) and part (b)).



19)

Solve each rational inequality and graph the solution set.

$x^2 + 4x + 3 \geq 0$

a) Graph the function $y = x^2 + 4x + 3$

b) The x-intercepts of the function $y = x^2 + 4x + 3$ are _____?_____

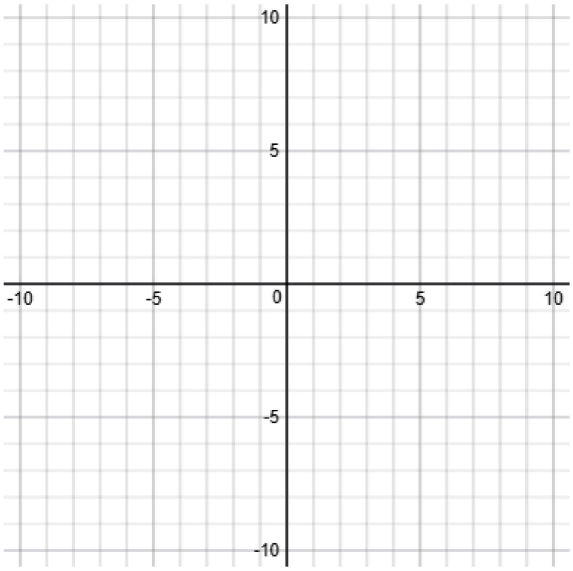
c) The interval on which the graph $y = x^2 + 4x + 3$ is above the x-axis is _____?_____

d) The solution set for the inequality $x^2 + 4x + 3 \geq 0$ is _____?_____

e) Is $x = -7$ a solution to the inequality $x^2 + 4x + 3 \geq 0$?

f) Is $x = 1$ a solution to the inequality $x^2 + 4x + 3 \geq 0$?

g) Is $x = 6$ a solution to the inequality $x^2 + 4x + 3 \geq 0$?



20)

Solve each rational inequality and graph the solution set.

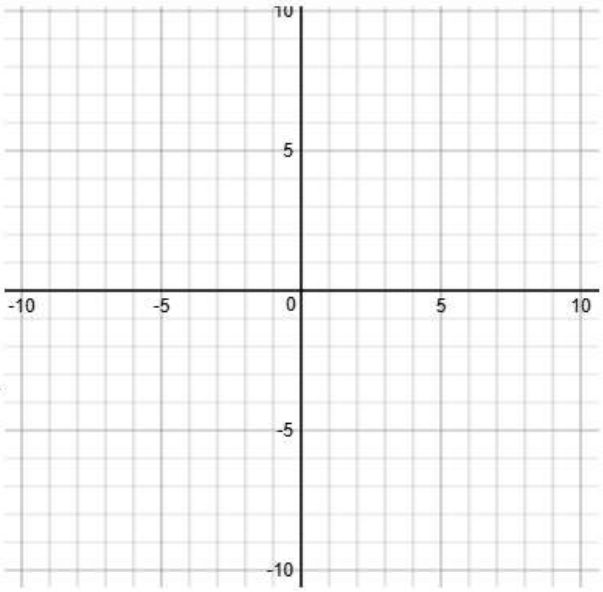
$4x^2 + 11x + 6 < 0$

a) Graph the function $y = 4x^2 + 11x + 6$

b) The x-intercepts of the function $y = 4x^2 + 11x + 6$ are ___?___

c) The interval on which the graph $y = 4x^2 + 11x + 6$ is below the x-axis is _____?_____

d) The solution set for the inequality $4x^2 + 11x + 6 < 0$ is ___?___



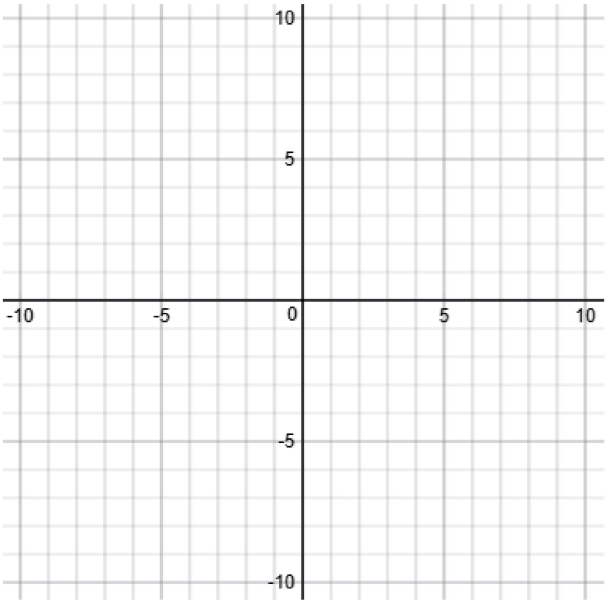
21)

Solve each rational inequality and graph the solution set.

$4x^2 - 5x \leq 6$

a) Graph the function $y = 4x^2 - 5x - 6$

b) The x-intercepts of the function $y = 4x^2 - 5x - 6$ are __?__

c) The interval on which the graph $y = 4x^2 - 5x - 6$ is below the x-axis is _____?_____

d) The solution set for the inequality $4x^2 - 5x \leq 6$ is __?__